observable is identified with a linear operator. We can already begin to see that an operator is a way of packaging up states along with their eigenvalues, which are the possible results of measuring those states. These ideas should become clear as we move forward.

Let's recall some important points from our earlier discussion of spins. First of all, the result of a measurement is generally statistically uncertain. However, for any given observable, there are particular states for which the result is absolutely certain. For example, if the spin-measuring apparatus $\mathcal{A}$ is oriented along the z axis, the state $|u\rangle$ always leads to the value $\sigma_z = +1$. Likewise, the state $|d\rangle$ never gives anything but $\sigma_z = -1$. Principle 1 gives us a new way to look at these facts. It implies that each observable (σ_x, σ_y, and σ_z) is identified with a specific linear operator in the two-dimensional space of states describing the spin.

When an observable is measured, the result is always a real number drawn from a set of possible results. For example, if the energy of an atom is measured, the result will be one of the established energy levels of the atom. For the familiar case of the spin, the possible values of any of the components are ±1. The apparatus never gives any other result. Principle 2 defines the relation between the operator representing an observable and the possible numerical outputs of the measurement. Namely, the result of a measurement is always one of the eigenvalues of the corresponding operator. Thus, each component of the spin operator must have two eigenvalues equal to ±1.[2]

Principle 3 is the most interesting. At least I find it so. It speaks of *unambiguously distinct states*, a key idea that we have already encountered. Two states are physically distinct if there is a measurement that can tell them apart without ambiguity. For example, $|u\rangle$ and $|d\rangle$ can be distinguished by measuring σ_z. If you are handed a spin and told that it is either in the state $|u\rangle$ or the state $|d\rangle$, to find out which of the two states is the right one, all you have to do is align $\mathcal{A}$ with the z axis and measure σ_z. There is no possibility